Dorotea Monaco

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WORK EXPERIENCE

DISEG - Politecnico di Torino - Italy, Turin

Website: https://luigigonnella.github.io/OpenSignalWebsite/ | **Link:** https://www.youtube.com/@OpenSignalPolito

Software Developer - OpenSignal Project

[03/01/2026 - 07/31/2026]

- Collaborated on the development of **OpenSignal**, a Python-based desktop application for seismic ground-motion analysis and selection.
- Contributed to the redesign and migration of the original MATLAB implementation to a modular Python architecture.
- Developed and integrated modules for ground-motion selection, response-spectrum analysis, spectral scaling and matching, and seismic disaggregation.
- Contributed to the integration of seismic databases and engineering models for ground-motion analysis.
- Developed and maintained the graphical user interface using **PySide6/QML**, with a focus on usability and cross-platform deployment.

UC Berkeley - Civil and Environmental Engineering Department - United States, Berkeley, California

Visiting Student Researcher

[09/01/2026 - 02/28/2027]

- Visiting Student Researcher at the **Department of Civil and Environmental Engineering, University of California, Berkeley**
- Conducting Master's thesis research at the intersection of **software engineering, machine learning, and structural engineering**
- Working on computational methods and software tools for structural and earthquake engineering applications.
- Collaborating with researchers on the development and evaluation of research-oriented computational workflows.

EDUCATION & TRAINING

Bachelor's Degree in Computer Engineering

Politecnico di Torino

City: Turin | **Country:** Italy

Final grade: 98 | **Level in EQF:** 6

Computer Science, Software Engineering, Computer Architecture, Operating Systems, Computer Networks, Databases, Algorithms and Data Structures, Embedded Systems, and Web Technologies.

Master's Degree in Computer Engineering

Politecnico di Torino [Current]

City: Turin | **Country:** Italy

Level in EQF: 7

Software Engineering, Machine Learning, Artificial Intelligence, Data Mining, Large Language Models, Computer Vision, Distributed Systems, Databases, and Software Architectures.

SKILLS

Programming Languages

JavaScript | C | Coding in Rust | TypeScript

AI & Machine Learning

Machine Learning (AI, DL, CNN) | Artficial Intelligence | Generative Models | Artificial Intelligence and Machine learning | Python (NumPy, Pandas, NLTK, Matplotlib, sklearn, TensorFlow)

ML Frameworks & Libraries

MLDL libraries PyTorch Numpy Pandas Scikit-learn Matplotlib | Natural Language Processing: transformers, sentence-transformers, HuggingFace, spacy, & NLTK | huggingface transformers | Python HuggingFace

Software Engineering & Backend

Web-based Software Development (REST JSON SQL PHP) | Git/Github, Docker, Gitlab | Databases (MongoDB MySql PostgresSql) | Python (computer programming) | NodeJs (ExpressJs) | Javascript (React.js) | Conoscenza di base di gestione ed utilizzo di WebSocket | Databases (PostgreSQL/PostGIS, MySQL, Redis)

PROJECTS

[05/01/2026 - 07/01/2026]

Virtual Staining of Histopathology Images Developed and evaluated deep-learning approaches for virtual H&E-to-IHC staining of histopathology images.

Implemented and compared GAN-based, deterministic, and latent diffusion architectures integrating pathology foundation models.

Designed patch- and region-level evaluation pipelines using perceptual, structural, and morphological metrics.

[12/01/2025 - 01/31/2026]

LLM Agents for Software Development Designed and evaluated LLM-based agentic workflows for software development and code-related tasks.

Explored different architectures and interaction strategies for integrating LLM agents into software engineering workflows.

Analyzed agent behavior and performance across code generation and development-oriented tasks.

[12/01/2025 - 02/01/2026]

Medical Image Augmentation with GANs Developed GAN-based models for synthetic medical image generation and data augmentation.

Implemented and compared DCGAN and conditional GAN architectures for controlled image generation.

Evaluated generated samples and their impact on downstream image classification using quantitative generative and classification metrics.

LANGUAGE SKILLS

Mother tongue(s): Italian

English

LISTENING: B2 **READING:** B2 **WRITING:** B2
SPOKEN PRODUCTION: B2
SPOKEN INTERACTION: B2